Heathkit HW-101 Transceiver — Restoration Plan

Rev. 2 — checked against the original Heath manual (595-1277-18)

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Safety. The HW-101 runs on **+700–850 V DC** from the HP-23 power supply (manual spec, p. 159). That voltage stays on the filter capacitors after the supply is unplugged. The manual (p. 117) says to connect **a lead from a good earth ground to the chassis GND terminal and to all test equipment** before power-up, and warns that lethal voltages are present above and below the chassis. Discharge before every hands-in session, and never operate the final cage with its cover off. Work one-handed when the set is live.

1. Source documents

Document	Where	Use
Heath HW-101 Manual 595-1277-18 (©1970; late printing, 245 pp. scan)	Supplied by owner; also on Latta's site	**Primary reference.** Page numbers below are the manual's printed numbers.
HW-101 Service Bulletins HW-101-1 … -85 (1971–79)	Latta site; -1 to -55 transcribed on the Heath list (Jul 1997)	Factory fixes. Several are already built into this late printing (§11).
HP-23 power supply manual	Latta site	Companion supply.
PS-23 AC Power Supply manual 595-2314 (©1979)	Supplied by owner	Supply section checked against it (Rev. 2.1).
W5RKL, *HW-101 and SB-10x Restoration and Troubleshooting*	w5rkl.com	First power-up and checks.
Latta (Frostburg) HW-101 restoration, alignment, HP-23 pages	frostburg.edu	A documented restoration with measured failures.
WB8JKR, *Optimizing the HW-101 / SB-100–102*; K4JPN, KM5Z notes	mikeyancey.com, nostalgickitscentral.com, pestingers.net	Improvements and tube-brand issues.
Heath list / Heathkit-Radios (groups.io)	puck.nether.net, groups.io	Field experience.

Numbering (schematic notes): 1–99 modulator board; 100s IF; 200s bandpass; 300s audio; 400s RF driver; 500s crystal; 600s heterodyne oscillator; 700s driver grid; 800s driver plate; 900s chassis. Resistors are 1/2 W unless marked. Manual voltages were measured with an 11 MΩ VTVM: ±10% (p. 144), ±20% on the transmitter chart.

2. Transceiver at a glance

2.1 Factory specifications (pp. 157–160)

Item	Spec
Coverage	3.5–4.0, **7.0–7.3**, 14.0–14.5, 21.0–21.5, 28.0–30.0 MHz (four 10 m segments)
Sensitivity	**< 0.3 µV** for 10 dB (S+N)/N, SSB
Selectivity	SSB 2.1 kHz @ −6 dB, 7 kHz @ −60 dB (3.395 MHz filter). Optional SBA-301-2 CW: 400 Hz / 2.0 kHz.

Item	Spec
Receiver audio	2 W at < 10% distortion
Transmitter input	180 W PEP SSB; 170 W CW (50% duty)
Output	**100 W on 80–15 m; 80 W on 10 m** into 50 Ω
Suppression	Carrier and unwanted sideband −45 dB; 3rd-order IMD −30 dB; harmonics −40 dB
Stability	< 100 Hz/hr **after 45 min warm-up**; < 100 Hz for ±10% line
Dial	5 kHz divisions; 35⅓ knob turns per 500 kHz
Complement	**20 tubes, 19 semiconductors** (p. 3); MPF-105 FET VFO, 2N3393 regulator
Power requirements	**700–850 V @ 250 mA (1% ripple); 300 V @ 150 mA; −115 V @ 10 mA; 12 V AC/DC @ 4.76 A**
Pilot lamps	**#47** (412-1), ×2 (parts list p. 45)
Weight / size	17½ lb; 14-13/16 × 6-5/16 × 13-3/8 in.

2.2 Tube complement (spec p. 159; V numbers from schematic, block diagram and troubleshooting chart)

V	Tube	Function	Known issues (source)
V1	6EA8	Speech amp (A) / cathode follower (B)	GE 6EA8s oscillate (~65 kHz) → VOX chatter, no carrier null (HW-101-21, -31). Manual: “very low output in USB/LSB → V1 defective” (p. 152 #33).
V2	6AU6	Isolation amplifier	—
V3	6AU6	1st IF amp (TX and RX)	Leakage → S-meter drift (HW-101-34). El-Menco draws excess current → can’t zero.
V4	6AU6	2nd IF amp	Brand-sensitive (HW-101-46).
V5	6EA8	1st transmit mixer (V5A)	Needs VFO injection (p. 152 #34).
V6	6CB6	2nd transmit mixer	Under shield.
V7	6CL6	Driver	**Weak V7 → transmit output falls off** (p. 153 #42).
V8, V9	**6146** (411-75)	Finals	Gassy 6146s → output falls off (p. 153 #42). Heath: 6146A only (HW-101-18, -39), later GE 6146B approved (HW-101-49). **Re-neutralize after any change.**
V10	6HS6	RF amplifier	**High failure.** Leakage → poor AGC (HW-101-24). A 6AU6 substitutes directly.
V11	6HS6	1st receive mixer	Same as V10; some 6HS6s had swapped pinouts (HW-101-12).
V12	6EA8	2nd receive mixer (A) / relay amp (B)	Defective V12 → relays stay energized (p. 153 #39; HW-101-23); relays won’t pull in TUNE (V12B, p. 152 #37).
V13	6BN8	Product detector (C) / AVC	Latta found a weak one.
V14	6GW8	Audio amp (A) / audio power (B)	Runs hot.
V15	6EA8	CW tone oscillator / amplifier	Shorts → sidetone faults, continuous keying (HW-101-74).
V16	12AU7	Carrier oscillator / BFO	Plate resistors R6/R7 drift (§3.2).

V	Tube	Function	Known issues (source)
V17	12AT7	VOX amp (A) / 100 kHz calibrator (B)	—
V18	0A2	150 V regulator	Fed through 7 W resistors R304/R305 (§3.2).
V19	12AT7	Heterodyne osc (A) / cathode follower (B)	Repeated failure: R212 220 → 330 Ω (HW-101-2).
V20	6AU6	VFO amplifier	**"Output low" → V20 weak** (VFO chart, p. 155). Its cathode feeds the FET's regulated supply → drift.

Brand note (K4JPN, W5RKL): El-Menco 6AU6, 6HS6, 6EA8, 12AT7, 12AU7 and 6CL6 tubes are repeatedly implicated. Keep spares of other brands.

2.3 Semiconductors (schematic diode/transistor tables; parts lists)

Designators	Type	Heath p/n	Use
D1, D2, D101, D201, D301, D902–D906	1N2071 (1 A, 600 PRV)	57-27	Switching, ALC rectifiers, anti-trip rectifiers, DC blocking
D901	1N4149	56-56	—
CR1–CR4	**FH-1100 hot-carrier** in this printing (1970 schematic table: 1N191)	56-87	Balanced modulator (HW-101-13 upgrade)
CR201, CR901, CR941	1N191 germanium	56-26	Calibrator harmonic generator, RF sampling, VFO shifter
D202	15 V zener (1N4166A)	56-25	Cathode bias
Q941	MPF-105 FET	417-169	VFO oscillator
Q942	2N3393	417-118	Voltage regulator (VFO supply)

Diode faults named in the manual:

- **D101 → slow receive recovery** (p. 153 #41).
- **D201 wrong type or backwards, D202 backwards or defective → no VOX** (p. 152 #38).
- **CR941 reversed → VFO shifter dead** (p. 155).
- **Q941/Q942 wired wrong → no VFO output** (p. 155).
- Bulletin HW-101-11's "D907" does not appear in this manual's diode table; check which diode it refers to on your unit.

3. What fails, ranked

3.1 Capacitors

All five electrolytics (parts lists pp. 12, 20, 24, 44; schematic):

Ref	Value (p/n)	Board / role	Replace with
C2	10 µF (25-115)	Modulator; V1A speech amp cathode bypass	10 µF 50 V
C12	20 µF (25-135)	Modulator; speech amp B+ decoupling (+300 V line via R14 4.7 kΩ)	22 µF 450 V
C212	10 µF (25-115)	Bandpass; VOX amp cathode bypass	10 µF 50 V
C304	20 µF (25-135)	Audio; audio-stage B+ decoupling	22 µF 450 V
C906	10 µF (25-147)	Chassis; **bias filter on the −130 V line (positive lead to ground)**	10 µF 250 V, **polarity reversed vs. B+ caps**

The manual does not print voltage ratings. The ratings above come from the circuit voltages (+300 V B+, −130 V bias). Keep audio and speech capacitances the same.

Other capacitor	Likelihood	Symptom	Action
C213 (VOX delay)	Medium	**Relays stay energized** (p. 153 #39C); VOX delay drift	Leakage-test; .47 µF per HW-101-63.
C946 (4700 pF, VFO)	Medium	Warble when tapped	Glue to the VFO wall (HW-101-42). Manual: separate the two **56 pF discs** by > 1/8 in. to cure VFO microphonics (p. 155).
C947 (VFO shift)	Medium	Intermittent shift	56 → 47 pF NPO (HW-101-19).
C208, C223, C604	Low–medium	No heterodyne injection (p. 152 #36)	Check value.
.005 µF 1.6 kV discs (×2, chassis)	Medium	HV bypass short	Replace with 2 kV+ parts.
.2 µF Mylars (27-34; modulator ×3, IF, bandpass, chassis ×2)	Medium	Leakage	Replace with film.
C701, C801 (680 pF)	Medium	R940 overheats	680 pF 500 V (HW-101-75).
Ceramic discs, micas	Low	—	Spot-test.

3.2 Resistors

These are 1/2–7 W carbon composition and wirewound parts, ~50 years old. Carbon comps drift high, worst where they run hot.

Ref	Manual value	Failure / symptom	Fix (source)
R106	**22 kΩ 1 W** (1-5-1, IF board; 1 W already in this printing)	Heats and changes → S-meter zero drift	22 kΩ 2 W wirewound (WB8JKR) or 33 kΩ 2 W (K4JPN). Keep all four connections on its pad.
R107	**100 kΩ 1 W** (1-28-1; HW-101-10 already incorporated)	S-meter drift	Check that it is the 1 W part.
R6, R7	**33 kΩ 1/2 W** (6-333; V16 plates)	Latta measured **330 kΩ** and 42 kΩ → low USB/CW gain	33 kΩ 1 W (HW-101-84); WB8JKR uses 27 kΩ.
R1	100 kΩ (V1A plate)	Heat damage	1 W (HW-101-84); also check R215 and R316.
R302	**3.3 kΩ 1 W** (bias series, −130 V → bias line)	Drift → wrong bias range	Measure; replace with 2 W metal film.
R304, R305	**1 kΩ 7 W, 2.5 kΩ 7 W** (0A2 / 150 V dropping)	Hot wirewounds; open → no 150 V (VFO, heterodyne oscillator)	Measure; check the surrounding board for heat damage.
R117	3.3 kΩ (AVC)	Open → AVC and S-meter decay too fast	Replace (HW-101-55).
R915	3.3 MΩ	Open → no SSB output, TUNE OK	Replace (HW-101-69).
R940, R904, R701	100 Ω / 1 kΩ	Overheat or short to the driver shield	Sleeve and dress (HW-101-37, -50; W5RKL).
R212	220 Ω	Repeated heterodyne-oscillator tube failure	330 Ω (HW-101-2).
R306, R307	(VFO shift network)	**VFO doesn't shift with the Mode switch** (p. 152 #35)	Check value.
R221	—	Not soldered at bandpass point F → **VFO output high** (p. 155)	Resolder.
R202	10 kΩ	*Early units only.* Limits drive	Jumper it (HW-101-14).
2.2 Ω 2 W (chassis)	—	—	Check.

Method: lift one lead, measure, and replace parts more than 10–20% out. Use carbon comp or carbon film in RF and oscillator circuits.

3.3 Tubes, by risk

1. **6HS6 (V10, V11):** the top HW-101 receive killer. Try a 6AU6 as a test substitute.
2. **6146 pair (V8, V9):** gas or emission ("output falls off", p. 153).
3. **6CL6 driver (V7):** weak → falling output.
4. **6EA8 (V1, V5, V12, V15):** oscillation, relay hang, sidetone.

5. **6AU6 (V3, V4, V20):** brand-sensitive; V20 weak → low VFO output.
6. **12AT7 (V17, V19), 12AU7 (V16), 6BN8, 6GW8, 0A2.**

Put each tube back in the socket it came from.

3.4 Mechanical, switches, relays

Item	Symptom	Fix (source)
RL2 contacts (carbon)	Chirp, slow recovery	Clean contacts 2 and 10; clean the relay base (HW-101-32).
RL1/RL2	Won't energize	V12B, relay or Mode wafer 2F (p. 152 #37).
Band/crystal switch wafers	Dead bands; no heterodyne injection	Manual: **"rotors reversed 180°" on the crystal or heterodyne-oscillator switch wafers** (p. 152 #36). A field unit had 3 of 5 wafers reversed.
Modulator and RF driver board hardware	**Transmitter unstable**	Tighten; **fit the lockwashers between the boards and chassis** (p. 153 #40B; HW-101-9). Also coil shield cover, tube-shield clips, and V8/V9 socket hardware (p. 153 #40).
SSB/CW filter slide switch	Low TX and RX	Clean (HW-101-53).
Driver preselector/load belts	Slip, won't track	Silicone 50/56/3 mm.
Preselector drums	Loose	Epoxy (Latta).
VFO / Jackson drive	Erratic, slips	Braid from the stop stud to the frame (HW-101-27); washers (HW-101-78). The VFO and dial come out as a unit (p. 154).
Comb brackets / detents	—	This printing already ships **steel comb brackets 204-2096** and **detents 266-1116**. Earlier units may need retrofitting (HW-101-8, -66).
Pilot lamps	Low output, S-meter drift	**#47 only** (manual p. 45; HW-101-20).

3.5 Calibrator and crystals

- **Supplied heterodyne crystals** (Y501–Y508) cover only the specified segments. Heath warns against other heterodyne crystals, because the driver coils are sequence-tuned in series-parallel (p. 120). A dead crystal gives no injection on that band (p. 152 #36B).
- **"Birdies" near 3740 kHz and 21,200 kHz are normal** (p. 120).
- Carrier crystals (modulator board): **3393.6, 3396.4, 3395.4 kHz** (404-205/-206/-215; p. 12). WB8JKR retrimmed drifted crystals: CW output went from 50 W to 110 W.
- Y201 100 kHz calibrator crystal: a choppy output causes spurs and noise, and can latch the rig in transmit through anti-VOX (HW-101-25; Latta).

3.6 HP-23 power supply (do this first)

Manual (p. 114):

- Set the **HP-23A's low-voltage switch to 300 V DC**. On the HP-23B, HP-23C and PS-23, make the internal connections for 300 V.
- **With the HW-101, the HP-23's own BIAS control is inoperative**; bias is set by the transceiver's BIAS control.
- With an **HP-13**, set the supply's bias control **fully clockwise** (p. 113).
- **Troubleshooting chart:** "relays stay energized → low B+, at 250 V instead of 300 V" (p. 153 #39E).

11-pin plug (Fig. 1-6):

Pin	Function
1	−130 V bias
2	Filament common
3	LV +300 V
4	HV +800 V
5	Ground (relay)
6	12.6 V filament
7	Ground (B− and bias)
8	No connection
9–10	Power switch
11	Relay N.O. (linear amplifier)

Typical recap (Latta, field):

Item	Original	Failure	Replace with
C1–C4	125 µF (500 V per the PS-23 manual)	Low capacitance	150 µF 500 V
C5 (B+)	40 µF	Open → CW tone, relay chatter (HW-101-73)	47 µF 450 V
C6, C7 (bias)	40 µF 150 V	Open	47 µF 160 V+
R1–R5 bleeders	100 kΩ 2 W	~150 kΩ	100 kΩ 2 W metal film
D1–D7	1N2071	Usually OK	1N4007
Line cord	2-wire	Safety	18/3 grounded (the PS-23 already has a 3-wire cord)

3.6.1 PS-23 supply (checked against the PS-23 manual, 595-2314, ©1979)

The PS-23 is the later (1979) kit supply in the HP-23 family. Page numbers are the PS-23 manual's printed pages.

How the PS-23 differs from the HP-23/HP-23A:

- **There is no front-panel B+ switch.** The low-voltage tap is chosen by an internal jumper at assembly: **350 VDC wiring** (Detail 5B, p. 10) or **275 VDC wiring** (Detail 5C, p. 11). On the schematic (illustration booklet p. 4) the brown transformer lead feeds 350 VDC and the brown-yellow lead feeds 275 VDC.
- **There is no bias control.** The PS-23 gives a fixed **−130 V** with no load (−100 V at 20 mA); bias is set on the transceiver.

- **A 2.92 A circuit breaker** (65-17) replaces the fuse. Primary current goes through the breaker and through pins 9–10, the AC switch in the equipment (p. 21).
- A **3-wire grounded line cord** is standard; the green wire goes to chassis lug F (p. 13).
- The output cable is a captive 8-wire cable with an 11-pin plug. The connector for the far end comes with the equipment (p. 17).

HW-101: LV tap. Needs **300 V @ 150 mA**, so the PS-23 must have the **350 VDC wiring** (Detail 5B: brown jumper from strip M lug 2 to capacitor K lug 2). Check that jumper before connecting. The low-tap wiring gives only about 250 V, and the HW-101 chart blames that for relays that stay energized (p. 153).

Specifications (p. 20):

Output	No load	Loaded	Duty
HV	820 V	700 V @ 250 mA (62.5 µF effective; < 1% ripple)	Continuous to 150 mA; 50% at 300 mA
LV high tap (350 VDC wiring)	350 V	300 V @ 150 mA (with 100 mA on HV)	Continuous to 175 mA
LV low tap (275 VDC wiring)	275 V	250 V @ 100 mA (with 100 mA on HV)	Continuous to 175 mA
Fixed bias	−130 V	−100 V @ 20 mA	—
Filament	—	12.6 VAC @ 5.5 A	—
Line	120/240 VAC, 50/60 Hz, 350 W		

Output pinout (schematic; Table 1, p. 17; Detail 7F, p. 14):

Pin	Cable wire	Function
1	Green	−130 V fixed bias
2	Black	Filament common
3	Orange	LV +350 V or +275 V (no load)
4	Yellow	HV +820 V (no load)
6	Red	12.6 VAC filament
7	White	Ground
9, 10	Brown, blue	AC switch loop in the equipment. If the equipment has no AC switching, jumper pins 9–10 at the plug and leave brown and blue unconnected (p. 14).
5, 8, 11	—	No connection

Recap list, with the PS-23 manual's values (parts list p. 3; schematic; circuit description p. 21):

Ref	Manual value	Function	Notes
C1, C2	125 µF 500 V (25-265 can)	HV voltage doubler	Replace both together; field practice is 150 µF 500 V.

Ref	Manual value	Function	Notes
C3, C4	125 µF 500 V (25-265 can)	LV half-wave doubler; C4 is also the pi-filter input	Same type of can as C1/C2.
C5	40 µF 450 V (25-36)	LV pi-filter output (with choke L1, 6 H)	An open C5 causes a CW tone and relay chatter (HW-101-73).
C6, C7	40 µF 150 V (25-20)	Bias filter	**Negative supply: the + lead goes to the ground lug** (p. 12).
R1–R4	100 kΩ 2 W	HV bleeders / equalizers across C1, C2	Drift high (field: ~150 kΩ); metal film 2 W.
R5	100 kΩ 2 W	LV bleeder	—
R7	1000 Ω 2 W	Bias filter series	Chart: no bias → D7, R7 or R8.
R8	27 kΩ 1 W	Bias bleeder	Field practice uprates to 2 W.
D1–D7	1N2071 (57-27)	D1–D4 HV doubler (two per leg); D5, D6 LV; D7 bias	1N4007 is the usual replacement.
L1	46-56, 6 H	LV filter choke	Chart: no LV → D5/D6, L1.
Breaker	2.92 A (65-17)	Primary protection	Opens on reset → transformer, rectifier/filter or load short.

Initial test (p. 15). Use a VTVM or VOM; a solid-state ohmmeter won't turn the diodes on.

- Find the meter polarity on R × 1 (about 10 Ω between strip H lugs 3 and 4). Put the correct lead on ground and switch to R × 10k.
- Socket readings: **pin 1 ≥ 22 kΩ; pins 3 and 4 ≥ 75 kΩ** (they rise slowly as the caps charge); **pin 7 = 0 Ω**; all others infinite.

Troubleshooting shortcuts (p. 19):

- **HV only half the normal value** → D1–D4 or C1/C2.
- **LV only half** → D5/D6.
- **All outputs half** → primary wired for 240 V on a 120 V line.
- **Dead** → breaker, the pins 9–10 loop, or the equipment's switch.

Caution (p. 17). The outputs are lethal and the large capacitors hold their charge. The manual allows shorting the cap lugs to chassis with a screwdriver blade. A **bleeder stick (resistor to ground)** is gentler on the caps and terminals. Work with one hand, on an insulating surface.

4. Restoration plan

4.1 Phase 0: Assess cold

- Photograph everything; note mods (RIT on V20, R202 present or jumpered, output connector changes, non-stock tubes).
- Pull and label every tube; test them; note brands.

- **Check switch-wafer rotor orientation**, especially the crystal and heterodyne-oscillator wafers.
- Inspect R304/R305 (7 W), R940/R701/R904, the final cage and RF choke, RL1/RL2, belts and drums.
- Check the receiver against the **resistance chart, Fig. 1-22** (fold-out from p. 155), with the power plug disconnected and all controls fully CCW. Sample values from the chart:

Point	Expected
V8/V9 plate	~95 kΩ
V18	34 kΩ
V19	38 kΩ / 9 kΩ
V17 pin	0 Ω with FUNCTION in CAL

4.2 Phase 1: HP-23 first

Recap per §3.6. Variac bring-up, unloaded, with test loads. Confirm about −130 V bias, **300 V** B+ and about 800 V HV. On a PS-23, check the 350 VDC wiring and run the socket resistance check (§3.6.1). Unloaded LV reads about 350 V and HV about 820 V.

4.3 Phase 2: Clean

- Vacuum and brush; cleaner on a rag or swab, never sprayed.
- Final cage: pipe-cleaner between the tuning-cap plates.
- DeoxIT every rotary switch, pot and slide switch (Function, Meter, SSB/CW filter).

4.4 Phase 3: Recap and resistor survey

1. Replace C2, C12, C212, C304 and C906 (C906 reversed polarity).
2. Replace the .005 µF 1.6 kV HV discs and the leaky Mylars.
3. Resistors per §3.2: R106 (2 W), R6/R7 (1 W), R1 (1 W), R302; check R304/R305, R117, R915, R212, R306/R307.
4. Clean RL2/RL1.
5. Fit #47 lamps.

4.5 Phase 4: Mechanical

Fit silicone belts, epoxy the drums, braid the VFO stop stud, glue C946 and separate the 56 pF discs. Tighten the board hardware **with lockwashers**.

4.6 Phase 5: VFO (pp. 154–155)

The VFO is a field-effect transistor oscillator (Q941) with a V20 6AU6 amplifier and Q942 2N3393 regulator, fed from the 0A2's +150 V.

Symptom (VFO chart)	Cause
Output low	V20 weak; or peak the top core of coil FM at dial 300
Output high	R221 not soldered at point F; VFO coil core turned to the wrong end (should be ~1¼ in. below the coil end)

Symptom (VFO chart)	Cause
No output	Q942 backwards; Q941 wired wrong; C950 lugs touching chassis; output cable shorted; C951 lead not through lug 2 to TP of L941
Shifter doesn't work	CR941 reversed
Microphonics	56 pF discs too close; long leads

Removing the VFO (p. 154):

1. Pull V19.
2. Remove the pilot-lamp screw.
3. Unplug the coax.
4. Unsolder bias, B+ and filament.
5. Loosen the Jackson-drive setscrews.
6. Remove the knob and the 3 and 9 o'clock screws.
7. Remove the spade-bolt nuts.
8. Work the VFO up and back.

When reconnecting: white-gray → BIAS, orange → B+, brown → FIL.

Field drift fixes: try several V20s; check Q942; coil 40-810 → 40-1976 (HW-101-1); remove RIT mods wired to the buffer.

4.7 Phase 6: Initial test (p. 117)

1. **Earth-ground the chassis and the test gear.**
2. Set the controls:

Control	Setting
FINAL TUNE	10 o'clock
LOAD	4 o'clock
MODE	LSB
FUNCTION	PTT
METER	PLATE
RF GAIN	Full CW
AF GAIN	PWR OFF
Others	Fully CCW
BIAS (right side)	**Fully CCW**

3. Connect an **8 Ω speaker; never run without a speaker or phones.**
4. Turn on. The pilot lamps should light, the filaments glow, and the meter read zero. Turn AF GAIN up until noise is heard. If there's no noise, check that the transmitter isn't keyed (PTT, key or TUNE).
5. Measure:

Point	Expected
Bandpass board point 5	**+275 V**
Audio board point 4	**−108 V** (bias)

Then compare against the **Receiver DC Voltage Chart (Fig. 1-20)** and **Transmitter Voltage Chart (Fig. 1-21)**, fold-out from p. 144.

4.8 Phase 7: Alignment (pp. 119–125)

Equipment:

- 11 MΩ VTVM (a 20 kΩ/V VOM loads the circuits).
- **50 Ω nonreactive dummy load rated 100 W; never light bulbs.**
- A receiver for WWV or a broadcast station on a 100 kHz multiple.
- A scope or SB-610 is recommended.

Do not transmit until directed (p. 119).

Presets (p. 119):

- CAL XTAL trimmer notch toward the 100 kHz crystal.
- DRIVER PRESELECTOR at 12 o'clock.
- MIC/CW fully CCW.
- LSB, BAND 3.5, dial 200, PTT, RF GAIN full CW, METER on ALC, AF GAIN at 9 o'clock.

Receiver

1. **S-meter:** with the antenna off and RF GAIN full CW, set ZERO ADJ (right side) for zero.
2. **Heterodyne oscillator (p. 120):**
   - VTVM (−DC) to **TP on the bandpass board near V19**. If it reads 0, use the adjacent lead of the 100 kΩ resistor.
   - "−.82 V or higher is normal."
   - On every band, adjust that band's heterodyne coil for **about −0.5 to −2 V**. Adjust in the direction that gives the **slower** change.
   - Coils 3.5, 14 and 28.5 are adjusted from the top of the RF driver board; the rest from the bottom (positions are marked on the shield cover).
   - Latta adds: don't exceed −2.0 V.
3. **VFO trimmers:** turn both trimmer screws (through the VFO chassis) CW until just snug, then ¼ turn CCW.
4. **CAL check:** FUNCTION CAL, BAND 3.5, rock the dial around 400. The signal should stop on VOX and peak with the preselector. Reset the preselector to 12 o'clock.
5. **IF:**
   - Peak **T201** for maximum volume.
   - Peak **T102** top and bottom slugs.
   - Turn **T103** slug *up* (CCW) **14 turns**, then peak (no more than ~1 turn).
   - Retouch all three.
   - Don't turn T201/T103 cores more than two turns; **never touch the sealed 6.8 MHz trap.**
6. **VFO (p. 121):**
   - Warm up ≥ **30 min**.
   - Find CAL near 4000 kHz (500), turn the dial to its CCW stop, and slip the circular dial so the "Stop" end meets the hairline (Detail 8-9A).
   - Walk the 4000 kHz CAL signal onto **500** with the **VFO COIL**.
   - Set the 3500 kHz CAL signal onto **0** with **both VFO trimmers**.
   - Repeat until both coincide. (VFO is 5000 kHz at "500", 5500 kHz at "0".)
7. **Driver grid and plate coils (pp. 121–122)**, in this order, peaking the S-meter:

Band	Preselector	Coils
3.5	12 o'clock	3.5
29.0	29.2 position	29

Band	Preselector	Coils
21.0	21.2	21 (the VFO harmonic sits near CAL; identify CAL with CAL/VOX)
14.0	14.2	14
7.0	—	7

Receiver benchmark (manual p. 122): at least S9+20 dB at 3700 kHz, decreasing to S3 at 29.2 MHz on the calibrator.

Transmitter (pp. 122–124; **coil cover must be on**)

1. **Neutralizing capacitor preset** (new build): through the hole in the RF cage front, turn CW until resistance is felt, then **one full turn CCW**.
2. **Bias:**
   - Set up: preselector 12 o'clock, MIC/CW CCW, FINAL at 10 o'clock, LOAD at 4 o'clock, LSB, 3.5, dial 200, PTT, METER on PLATE.
   - **Press PTT and set BIAS so the needle sits on the ▼ mark (above the "3") = 50 mA resting cathode current.** Press PTT only a few seconds at a time until set.
   - Check that REL PWR reads 0.
3. **T1:**
   - TUNE, raise MIC/CW for an indication, peak the preselector and FINAL.
   - Set MIC/CW for **≤ S3**.
   - Turn T1's slug **up 14 turns**, then peak (≤ ~1 turn).
   - Reduce MIC/CW and re-peak T1.
4. **Preliminary neutralization (3.5 MHz):**
   - In TUNE, find the FINAL setting for minimum plate current and the one for maximum REL PWR.
   - Adjust the neutralizing capacitor a small amount at a time until they **coincide**.
   - Run full power only briefly; **cool ≥ 30 s** in LSB between tries.
5. **Carrier null:**
   - MIC/CW fully CCW. In LSB with PTT, null with the CARRIER NULL **control**, then the CARRIER NULL **capacitor**.
   - Repeat in USB, and repeat both until the residual is about equal on LSB and USB.
   - With an 11 MΩ VTVM and RF probe at the ANTENNA jack center pin, aim for **¼ V or less**.
6. **Heterodyne coils for drive:**
   - TUNE, REL PWR.
   - Peak the preselector, FINAL and LOAD; set MIC/CW for **3–9**.
   - Peak HET OSC coil 3.5 on the **slow side**, then each band's coil.
7. **Driver neutralization (21.2 MHz):** peak, then rock the preselector. If peaking is ragged, reposition or bend the **white insulated driver-neutralizing wire in hole W** of the RF driver board.
8. **Final neutralization (14.2 MHz):** peak, then adjust until maximum REL PWR and minimum plate current coincide.
9. **ALC zero:**
   - Re-zero the ALC position while receiving at 29.5, then check zero on every band.
   - If a band isn't zero, retouch that band's heterodyne coil.
   - The meter resting below zero on transmit is normal.
10. **Crystal calibrator (p. 124):**
    - Antenna wire to an external receiver tuned to WWV.
    - FUNCTION CAL, AF GAIN CCW. Adjust the **CAL XTAL trimmer** (bandpass board) for zero beat.
    - **Use 15 MHz WWV:** a 20 Hz error at 100 kHz becomes 740 Hz at 3.7 MHz.

11. **VFO shifter (p. 125):** 3.5, dial 200, CAL. Zero beat in **USB**, switch to **LSB** without moving the dial, and adjust **SHIFT ADJUST** on the VFO. Recheck USB.
12. **Dial calibration (p. 125):** zero beat the calibrator at 3.5 MHz / dial 0. Hold **ZERO SET** and turn the knob to bring 0 under the index. Verify on a known station (e.g. CHU 7335 kHz).

4.9 Phase 8: Operating tune-up (pp. 140–141): every session, and after any ±20 kHz move

1. Set the band and frequency. METER on PLATE, MIC/CW fully CCW, 50–75 Ω load.
2. **TUNE:** the meter should read **50 mA (▼)**. If it doesn't, redo the bias.
3. METER on REL PWR, LOAD at 4 o'clock, FINAL to the band's mark (Fig. 1-19).
4. Raise MIC/CW slightly. Peak the PRESELECTOR, FINAL and LOAD.
5. Raise MIC/CW until the output stops increasing, then re-peak.
6. METER on PLATE: about **"40" on the scale = 250 mA**. (Scale: 3 = 50, 6 = 100, 9 = 150, 20 = 200, 40 = 250, 60 = 300 mA.)
7. **Check:** maximum REL PWR must fall at the same FINAL setting as the plate dip. **If it doesn't, do not operate; re-neutralize.**
8. Return MIC/CW to CCW.

Limits:

- **Full output no longer than 30 s at a time; cool ≥ 1 min.**
- SSB (PTT, METER on ALC): set MIC/CW so voice peaks reach about **S3, keeping below S6** on the ALC scale.

4.10 Phase 9: Optional improvements (reversible)

Mod	Change (source)
Stable meter zero on TX	10 MΩ from meter-zero pot pin 3 to T102 pin 2 (WB8JKR)
Carrier crystals on frequency	Series/parallel trim caps (WB8JKR)
More BFO injection	C17 12 → 100 pF; R6/R7 → 27 kΩ (WB8JKR)
Less raspy audio	C119 500 pF → .001 µF; C306 → .01 µF (WB8JKR)
Mic low end	C11 .001 → .01 µF (WB8JKR)
Faster CW relays	R328 470 kΩ → 1 kΩ (WB8JKR)
Sidetone	R326 → 2.2 or 3.3 MΩ; volume control per HW-101-70
Birdies	.01 µF from the common filament point to ground
Avoid	Replacing the RCA antenna jack with SO-239; use an adapter (Latta)

5. Symptom → cause quick reference (manual chart pp. 146–155 + bulletins)

Symptom	Manual chart / bulletins
Very low USB/LSB output	T1 not aligned; low mic level; V1 (p. 152 #33)
No VFO injection at V5/V12	V20; power (p. 152 #34)

Symptom	Manual chart / bulletins
VFO won’t shift with mode	Mode wafer 1F; R306/R307 (p. 152 #35); CR941 (p. 155)
No heterodyne injection at V6/V11	Crystal Y501–Y508, coax, C208/C223/C604, V19, L601–L608, no 150 V, reversed wafer rotors (p. 152 #36)
Relays won’t energize (TUNE / VOX)	V12B, RL1/RL2, wafer 2F; VOX SENS, V17A, D201, D202, anti-trip (p. 152 #37–38)
Relays stay energized	PTT/key closed, **C213**, **V12**, **B+ at 250 V not 300 V** (p. 153 #39); -23
Transmitter unstable	Neutralization, **board lockwashers**, L801–L805, coil cover, tube-shield clips, V8/V9 hardware, lead lengths (p. 153 #40); -9
Slow receive recovery	**D101** (p. 153 #41); RL2 carbon (-32)
Output falls off	Heat, bias, load, **gassy 6146s, weak V7** (p. 153 #42)
Dial zero varies band to band	Calibrator not exactly 100 kHz; heterodyne coils; VFO alignment (p. 153 #43)
Low output, all modes	IF-board wire not in hole 3 on the bandpass board (p. 153 #44)
S-meter drift / won’t zero	R106, R107 (-10), V3 leakage (-34), #47 lamps (-20), -67
Poor AGC	6HS6 leakage (-24); R117 (-55)
Carrier suppression	-3, -28, -38

6. Operation notes

- 50–75 Ω nonreactive load always (manual p. 139). HP-23 on **300 V**.
- Earth-ground the chassis.
- Follow the 10-step tune-up every time; check that peak and dip coincide.
- Leave ventilation clear; don’t stack equipment on the rig (p. 131).

7. Tips and tricks

- Label tubes by socket; tube brand matters (6AU6, 6HS6, 6EA8).
- Photograph wafer and rotor orientation before touching switches.
- Grid-dip heterodyne coils to find a dead crystal fast.
- Spares worth keeping: 6HS6 ×2, 6EA8, non-El-Menco 6AU6s, 6CL6, 12AT7, a matched 6146 pair, #47 lamps, silicone belts.

8. Maintenance schedule

Interval	Task
Each session	45 min warm-up for the stability spec (30 min minimum); calibrator zero-set; 10-step tune-up.
6 months	Exercise switches and pots; S-meter zero.
Yearly	Bias (50 mA ▼); carrier null; receive benchmark (≥ S9+20 at 3.7 MHz); relay cleaning if needed.

Interval	Task
2–3 years	Neutralization check at 3.5 and 14.2; retighten board hardware and lockwashers; DeoxIT.
5 years	Tube test (6HS6, 6146, 6CL6 first); belts; HP-23 check.
~15–20 years	Re-check electrolytics.

9. Parts sources

- Mouser / Digi-Key: electrolytics, 2 W metal film, NPO caps, 1N4007, 2 kV discs.
- HP-23 rebuild kits (HP-23RL and similar); silicone belts; Heathkit-Radios (groups.io) for crystals, 6HS6s and boards.

10. Sources

- Heath, *Heathkit Manual for the SSB Transceiver Model HW-101*, 595-1277-18 (©1970).
- Heath service bulletins HW-101-1 … -85 (Latta site; Heath list, Jul 1997).
- W5RKL, *HW-101 and SB-10x Restoration and Troubleshooting*.
- Heath, *Heathkit Manual for the AC Power Supply Model PS-23*, 595-2314 (©1979).
- Latta, Frostburg State HW-101 pages (restoration, alignment, modifications, HP-23).
- WB8JKR, *Optimizing the HW-101, SB100-102*; K4JPN, *Improve Your HW-101*; KM5Z refurb summary (pestingers.net).
- Heath list threads: “Problems with HW-101 — a possible solution” (Dec 2020); “Drifting HW-101” (Jun 2007); “S meter drift” (Feb 2003).

11. Rev. 2 change log (manual review)

Item	Rev. 1	Rev. 2 (per manual)
Power requirements	HP-23 outputs only	Spec: **700–850 V @ 250 mA, 300 V @ 150 mA, −115 V @ 10 mA, 12 V @ 4.76 A** (p. 159)
HP-23 bias	Not addressed	HP-23 BIAS control inoperative with the HW-101; set by the transceiver’s BIAS. HP-13 bias fully CW (pp. 113–114).
Sensitivity, output, stability	Not stated / 100 W	< 0.3 µV; 100 W on 80–15, 80 W on 10; < 100 Hz/hr after **45 min** (pp. 157–158)
40 m coverage	7.0–7.5 implied	**7.0–7.3 MHz**
V-number table	“Verify”	Confirmed from schematic, block diagram and troubleshooting chart
Diodes	Partial, CR201/CR901 only	Full table; CR1–4 are FH-1100 in this printing; D202 15 V zener; D907 not in the manual’s table

Item	Rev. 1	Rev. 2 (per manual)
Electrolytics	Ratings guessed (15 V / 350 V)	Five confirmed with p/n; manual prints no voltages; C906 is the **−130 V bias filter**
R106 / R107	22 kΩ 1 W / 100 kΩ ½ W	22 kΩ 1 W / **100 kΩ 1 W already** in this printing
Power resistors	—	Added R302 3.3 kΩ 1 W (bias), **R304 1 kΩ 7 W / R305 2.5 kΩ 7 W** (0A2 dropping)
Bias	“50 mA dot” vs. Latta “3” (conflict)	**▼ mark above “3” = 50 mA resting cathode current**, set in PTT; checked in TUNE before every tune-up (pp. 122, 140)
Loading	“~40”	“40” = **250 mA**; full meter scale table (p. 139)
Initial test	Generic	Manual Initial Test settings and **+275 V / −108 V** checkpoints; earth-ground warning (p. 117)
Alignment	Field summary	Full manual sequence: T103 and T1 “up 14 turns then peak”, VFO trimmer preset, neutralizing preset one turn CCW, ALC re-zero at 29.5, calibrator at 15 MHz WWV
Receive benchmark	S9+20 → S6 (W5RKL)	Manual: **≥ S9+20 at 3700 kHz → S3 at 29.2 MHz**
Troubleshooting	Bulletins only	Added manual chart items (D101, C213, V7, R221, CR941, rotors 180°, 250 V B+) and VFO chart
Late-printing features	—	Steel comb brackets, new detents, #47 lamps, FH-1100 modulator diodes already supplied
Supply (Rev. 2.1)	HP-23 field recap; “PS-23: internal 300 V connection”	Added §3.6.1, checked against the **PS-23 manual**: no B+ switch (350/275 VDC set by internal jumper), fixed −130 V bias with no control, 2.92 A breaker, 3-wire cord, full recap table with designators, pinout, initial resistance test, troubleshooting. C1–C4 rating corrected to **500 V**.